CRISPR in action
practice and theory of genome engineering

Day 4

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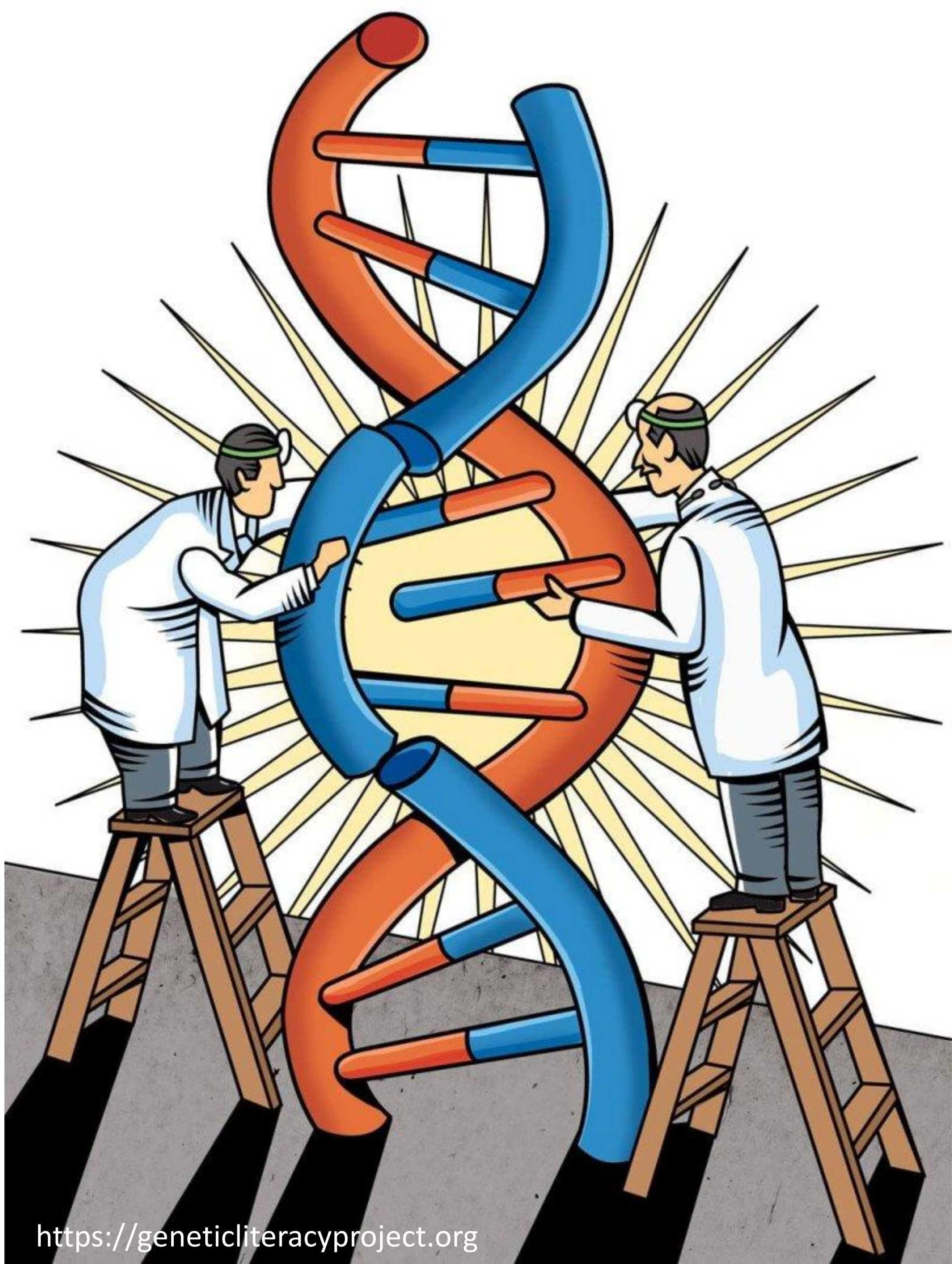


Genotyping approaches

PCR fragment analysis (gel/fragment size)
- Detects deletions/insertions by amplicon size shift

Sanger sequencing
- Confirms precise base edits or small indels
- Requires clean primer placement for good reads

Amplicon sequencing (NGS)
- Sensitive detection of mosaicism or low-frequency edits/genotypes
- Quantitative allele distribution
- Requires short amplicons (NGS read length)

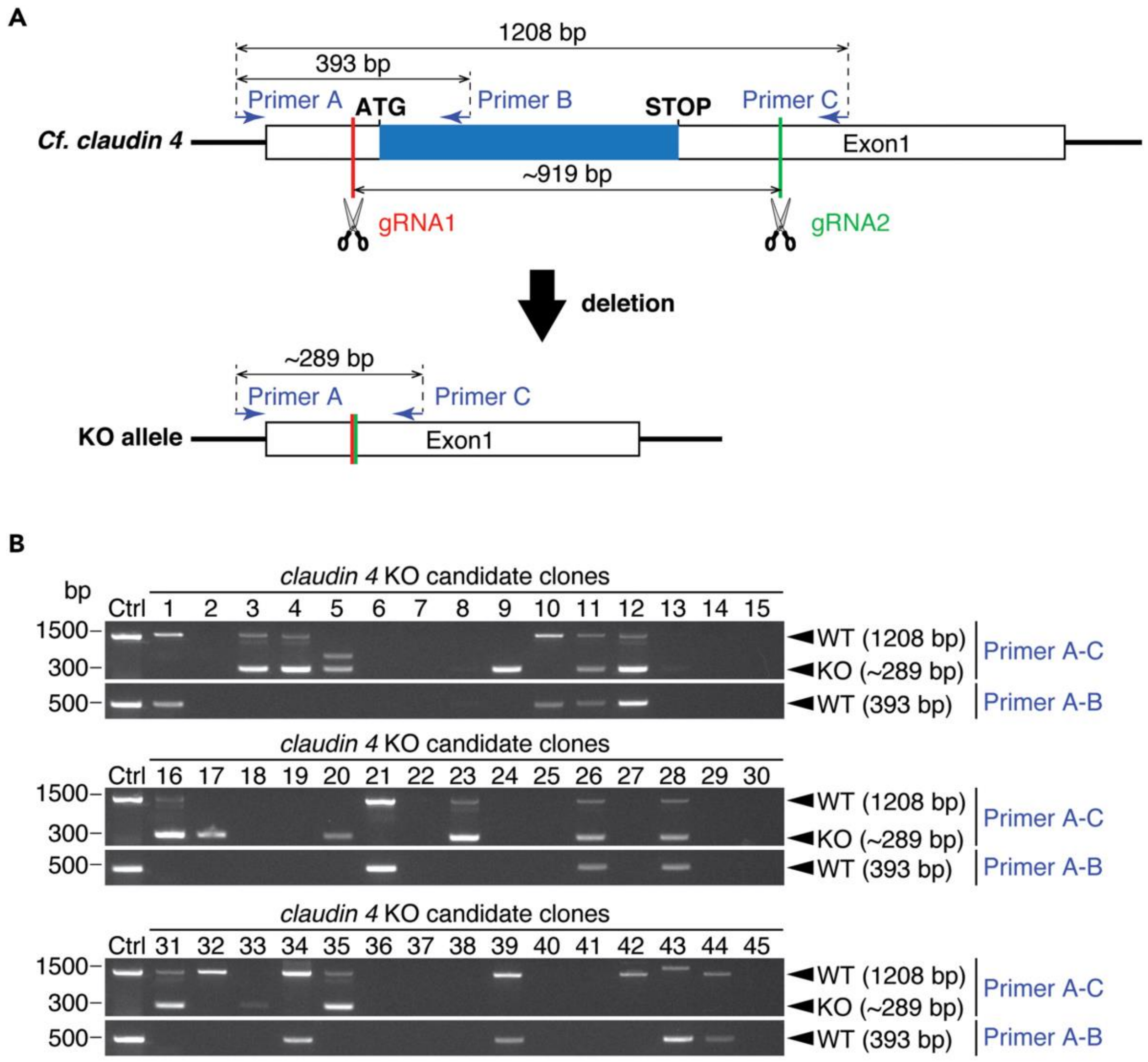
PCR fragment analysis

Primers should be able to amplify wild-type and edited alleles

Shorter genes/loci: one PCR, size difference
Longer genes/loci: two PCRs, presence/absence of bands

Design good PCR primers (manual, primer3 etc):

- 18–24 bp, GC 40–60%, Tm 55 - 65 °C
- Avoid dimers, hairpins, repetitive bases
- G or C at 3′ end
- Check primer uniqueness (BLAST)



https://doi.org/10.1016/j.xpro.2024.103179

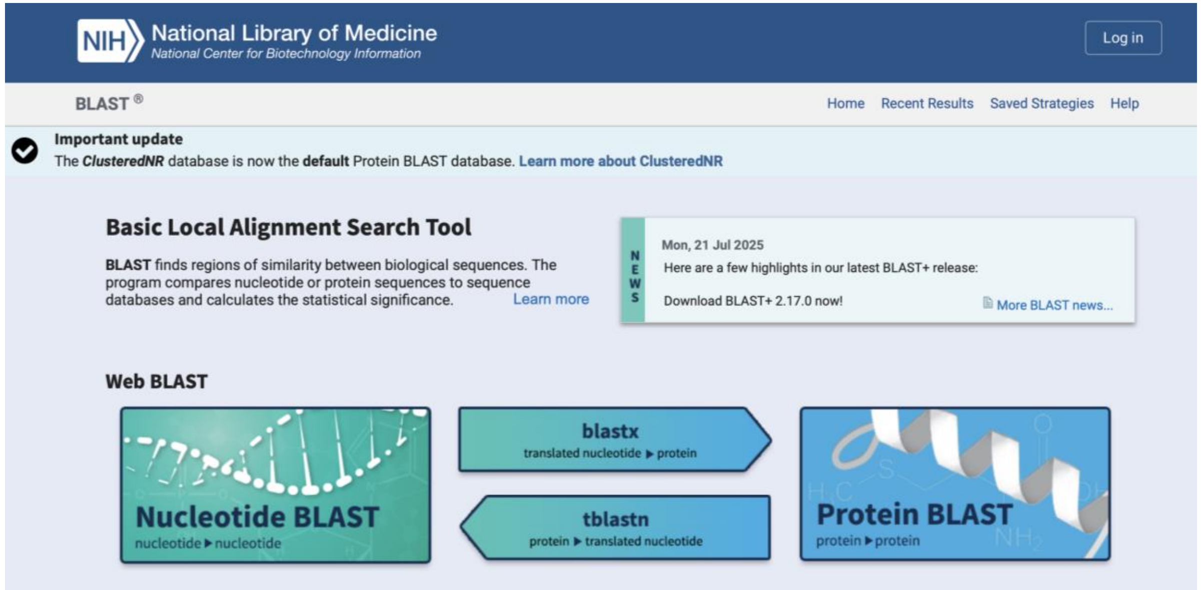
BLAST options

NCBI-BLAST

Organism-specific database

Primer-BLAST
https://www.ncbi.nlm.nih.gov/tools/primer-blast/

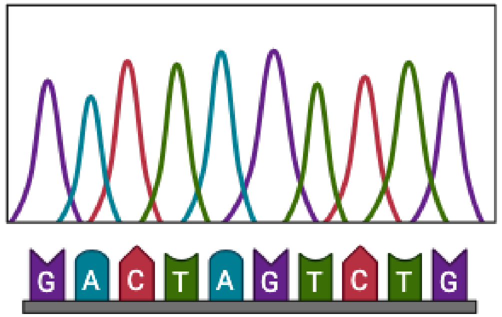
BLAST+ (command line)
https://conmeehan.github.io/blast+tutorial.html



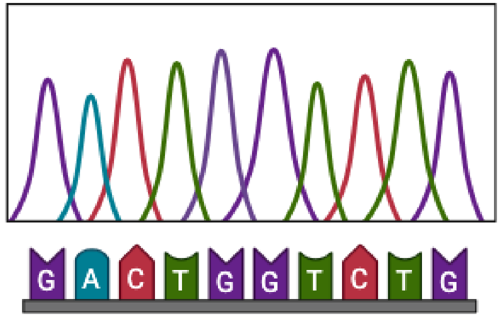
Sanger

- Short indels, precision edits
- Amplicons preferentially <900 bp length
- Edit > 30 and < 850 bp from primer
- Avoid repetitive or homopolymer runs in amplicon, if unavoidable try to avoid in sequencing (use fwd, rev primer)
- Single clean band essential

Design good PCR/sequencing primers!



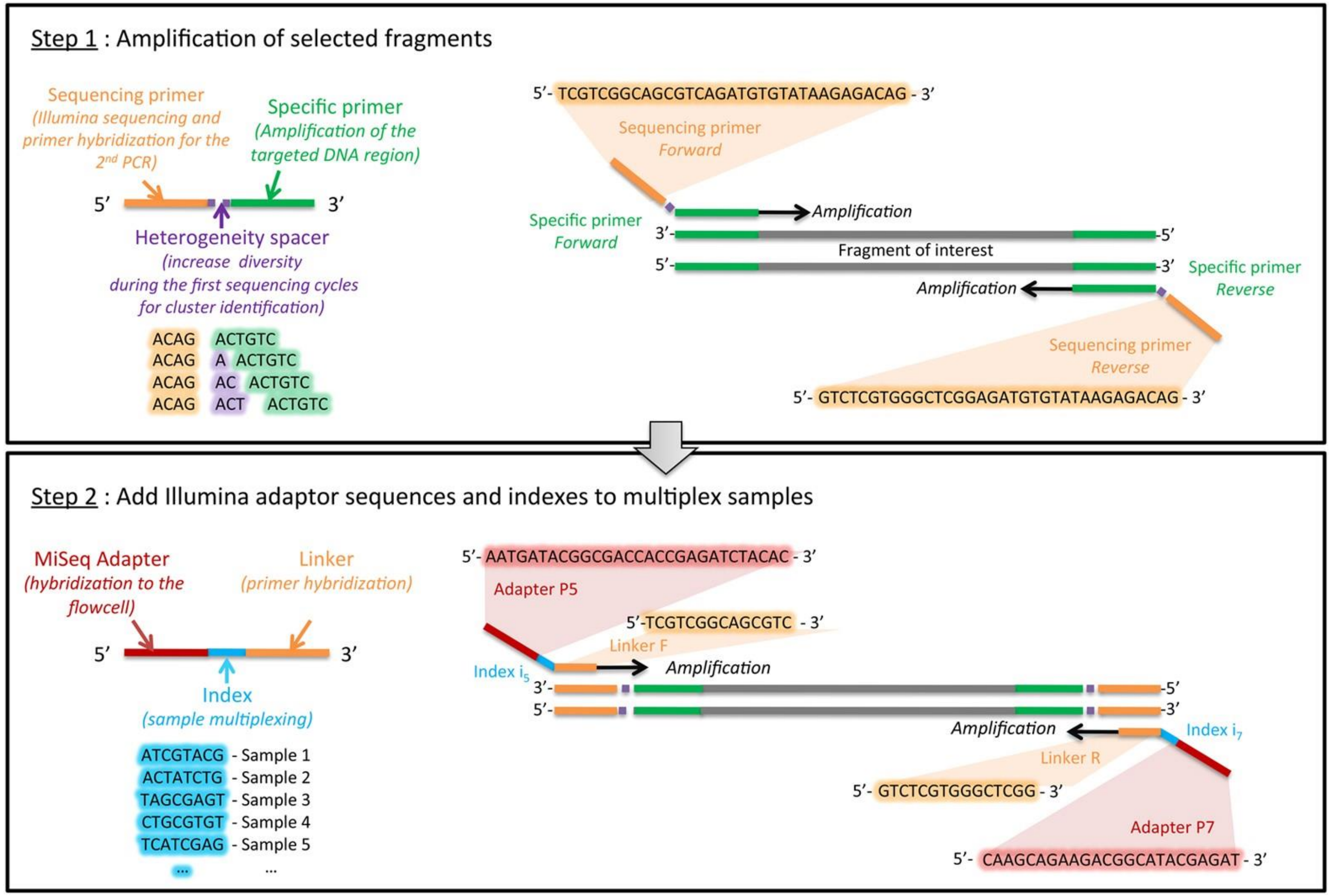
Wild-type



Edit (A to G)

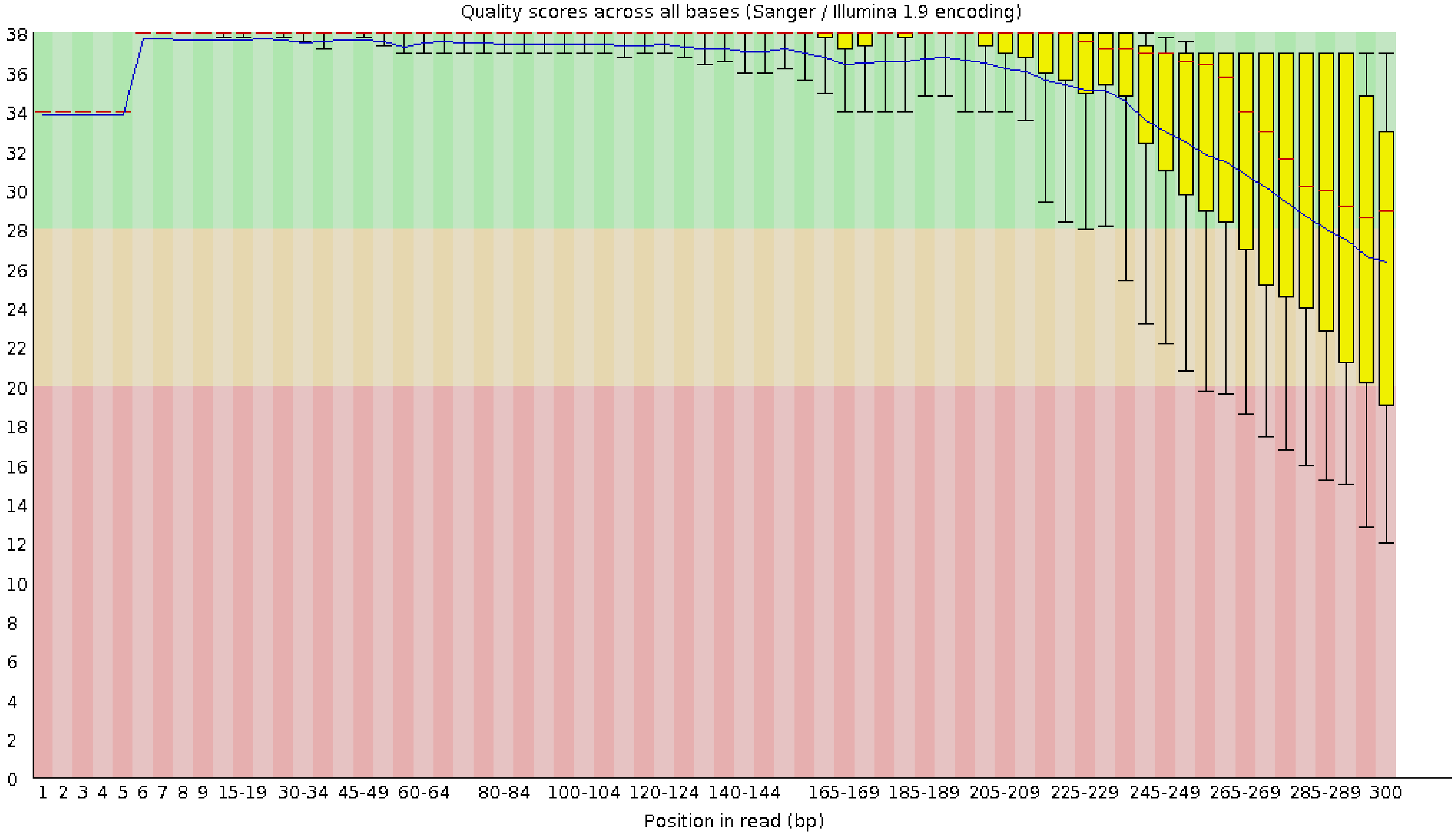
Amplicon-seq

- Allele distributions, quantitative
- Amplicon length constrained by NGS read length
- Short-read: 150 PE ideal, amplicon 180 – 230 bp (max. 270 bp), edit in middle → consensus reads
- Multiple bands tolerated but not ideal (waste reads)
- Design good PCR/sequencing primers (primer3 + BLAST for design)
- Two PCRs: target specific primers, Illumina compatible primers + adapters

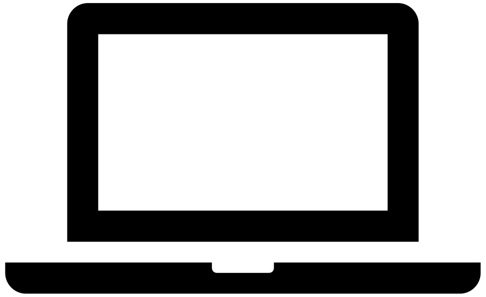


https://doi.org/10.1038/srep41948

Sequence quality dropoff



https://www.ecseq.com/support/ngs/why-does-the-sequence-quality-decrease-over-the-read-in-illumina



Go to *galK* file on Benchling and design primers for genotyping your PTC and KO